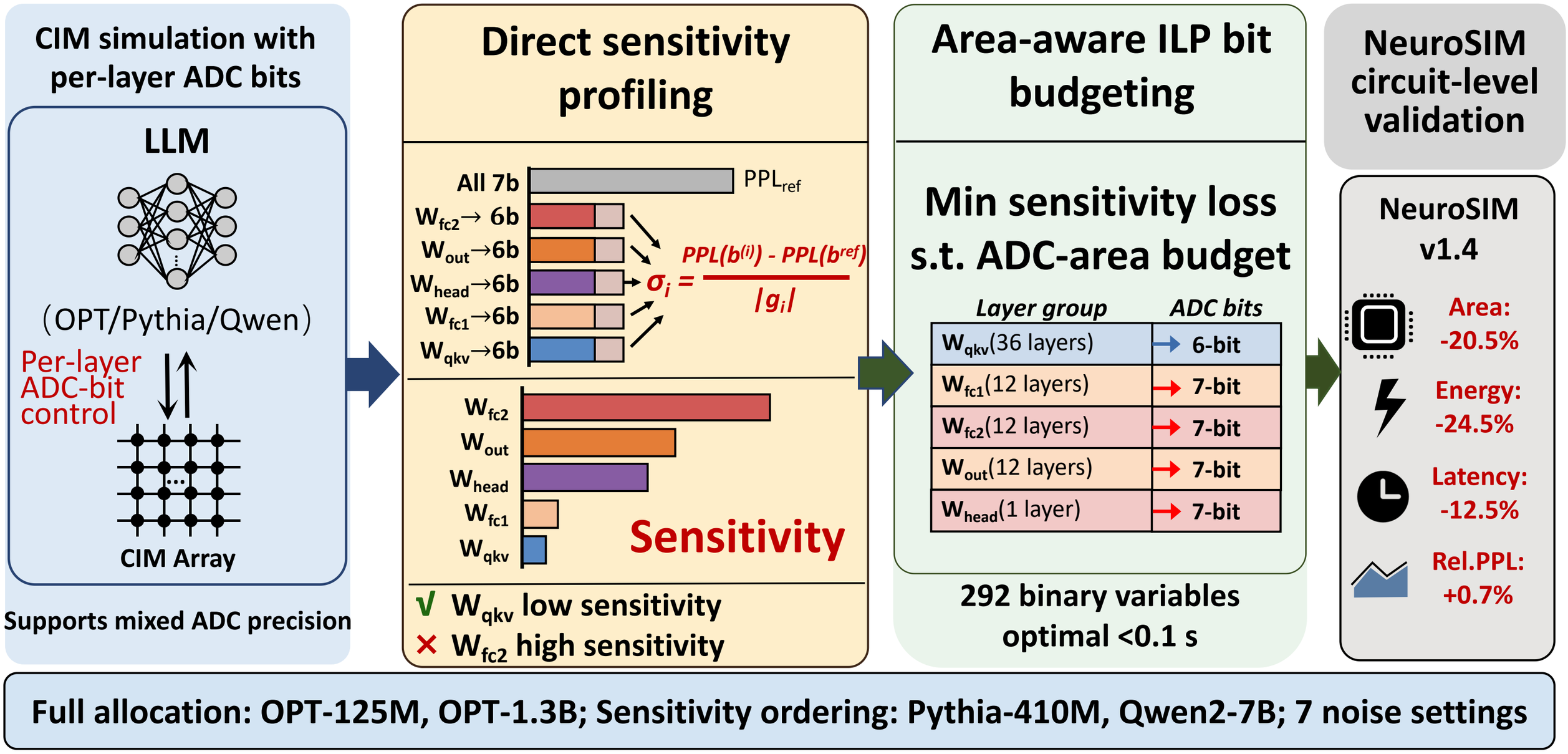
CIM simulation with per-layer ADC bits
LLM
(OPT/Pythia/Qwen)
Per-layer ADC-bit control
CIM Array
Supports mixed ADC precision
Direct sensitivity profiling
All 7b
PPL_ref
W_fc2→ 6b
W_out→6b
W_head→6b
W_fc1→6b
W_qkv→6b
σ_i = (PPL(b^(i)) - PPL(b^ref)) / |g_i|
W_fc2
W_out
W_head
W_fc1
W_qkv
Sensitivity
√ W_qkv low sensitivity
× W_fc2 high sensitivity
Area-aware ILP bit budgeting
Min sensitivity loss s.t. ADC-area budget
Layer group
ADC bits
W_qkv(36 layers) → 6-bit
W_fc1(12 layers) → 7-bit
W_fc2(12 layers) → 7-bit
W_out(12 layers) → 7-bit
W_head(1 layer) → 7-bit
292 binary variables
optimal <0.1 s
NeuroSIM circuit-level validation
NeuroSIM v1.4
Area: -20.5%
Energy: -24.5%
Latency: -12.5%
Rel.PPL: +0.7%
Full allocation: OPT-125M, OPT-1.3B; Sensitivity ordering: Pythia-410M, Qwen2-7B; 7 noise settings